SuPAR AR Inspection — Technology Analysis

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Sources: trilion.com, supar.eu, cdmvision.com, Apple App Store, Europac3D, Metrology News, HoloCode

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1. Executive Summary

SuPAR (Augmented Reality Interactive Inspection) is a mobile AR platform that overlays CAD geometry onto physical parts in real-time for visual quality inspection. Developed by **CDMVision** (Istanbul, Turkey) and distributed internationally by **Trilion Quality Systems** (USA), **Europac3D** (UK), **APM Technologies** (India), and **pmargage** (EU).

The system claims to reduce inspection time by **75–80%** compared to traditional 2D drawing-based methods. It targets automotive, aerospace, defense, heavy industry, and general manufacturing.

Verdict: A well-positioned niche tool for *visual* quality checks (presence/absence, gross misalignment) on the shop floor. Not a metrology instrument — it lacks quantified accuracy tolerances, does not replace CMMs or 3D scanners, and its precision is fundamentally limited by consumer-grade AR hardware. Strong for rapid go/no-go checks; insufficient for dimensional inspection.

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2. Company & Distribution

Entity	Role	Location
CDMVision (CDMVISION Yazılım Hizmetleri Ticaret A.Ş.)	Developer / OEM	Pendik, Istanbul, Turkey + Fellbach, Germany
Trilion Quality Systems	North American distributor	USA
Europac3D	UK / Ireland distributor	UK
APM Technologies	India distributor	India

- CDMVision has **20+ years** experience in AR/VR application development
- Other products: **HoloZee** (AR/VR experiences), **CAD2AR Framework** (CAD-to-AR conversion pipeline)
- Notable clients: Arçelik, Havelsan, Türk Telekom, Anadolu ISUZU, Stuttgart 21
- Award: **TITAN Awards Gold Winner** — Best AR/VR Technology Innovation

Trilion's main business is **Digital Image Correlation (DIC)** and optical measurement (GOM/ZEISS ecosystem). SuPAR is sold alongside ATOS Q (3D scanning), TRITOP (optical CMM), and ZEISS INSPECT. This positions SuPAR as a complementary rapid-check tool within a full metrology stack — not a replacement for high-precision instruments.

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3. Software Architecture

SuPAR is a **three-component system**:

3.1 SuPAR Composer (Desktop)

Purpose: CAD data preparation for AR inspection.

Feature	Description
Tessellation of edges	Converts CAD model edges into tracking geometry
Inspection checkpoint creation	Define named inspection points on the 3D model
Complex assembly support	Partial/regional inspection of large assemblies
Physical-digital mock-up creation	Build AR overlays for assembly guides
3D report viewer	Review completed inspection reports in 3D
Polyline/polygon markup	Define inspection features beyond CAD geometry (weld seams, sealant paths)

Platform: PC (Windows implied, not explicitly stated).

Critical gap: Supported CAD import formats are never disclosed — a significant omission for any industrial tool. Common formats (STEP, IGES, JT, CATIA, NX) would be expected, but none are confirmed.

3.2 SuPAR App (Mobile)

Purpose: On-site AR inspection and documentation.

Attribute	Detail
iOS	iPhone 6S+, iPad 5th gen+, iPod Touch 7th gen. Requires iOS 13.0+
Android	Available on Google Play (com.cdm.supar)
App size	~530 MB (iOS)
Price	Free download, **$249.99** in-app purchase (Standard tier)
Latest version	2025.1.138 (Feb 4, 2025)

Key features:

- Real-time CAD overlay via device camera
- Interactive AR tracking with image comparison
- Detection of missing, excess, or misaligned components
- Photo + comment annotation at inspection points
- Report generation (PDF, Excel, DOCX, 3D format)
- Password-protected reports
- Assembly guide mode (physical-digital mock-up)
- Video conferencing module
- Offline capability
- Multi-language support
- "Smartpoints" AI integration
- Stage system for complex multi-step inspections
- Draft label addition for inspection templates

3.3 SuPAR Web Viewer (Cloud)

Purpose: Free browser-based 3D inspection report viewer.

- No installation required
- Mobile-compatible
- Annotation on 3D models
- Unlicensed/free access for report consumers

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4. Core Technologies — Deep Dive

4.1 AR Tracking

Method: Model-based markerless tracking.

The system creates **tracking geometry from CAD data** (via Composer's edge tessellation) and performs real-time alignment between the rendered CAD edges and the physical part's actual edges as seen through the device camera.

Aspect	What's known	What's missing
Tracking type	Markerless, model-based	No detail on specific algorithm (edge-based, feature-based, hybrid)
Initialization	"Snap-on" to real parts	No detail on initial pose estimation
Platform APIs	Uses device camera for AR	No confirmation of ARKit/ARCore dependency
LiDAR usage	Europac3D mentions "3D scanning technology built into latest Apple devices"	Not confirmed whether LiDAR is *required* or simply *leveraged when available*
Stability	"Smoother visualization, increased stability" (2026 release)	No numerical tracking error data

Critical analysis: The app supports iPhone 6S (no LiDAR) and iPad 5th gen (no LiDAR), so LiDAR is clearly **not required**. When available (iPad Pro, iPhone 12 Pro+), it likely enhances depth estimation and initial alignment. The core tracking is almost certainly vision-based edge matching — the Composer's "edge tessellation" creates the reference geometry that the app matches against camera frames.

Probable technology stack: ARKit (iOS) / ARCore (Android) for device pose estimation, with a proprietary model-based tracking layer on top for CAD-to-real alignment. CDMVision's CAD2AR Framework (mentioned on their website) is likely the underlying engine.

4.2 Edge Detection & Processing

- "Enhanced edge creation" in Composer (2026 release)
- "Dynamic edge generation" — edges computed on-the-fly in the app
- "Advanced search and filtering capabilities" for edge management
- "Existing edge image processing in AR" — likely Canny/Sobel-type real-time edge detection on camera frames, matched against projected CAD edges

This is the heart of the system: project CAD model wireframe → extract edges from live camera → minimize alignment error between projected and detected edges.

4.3 AI Module ("Smartpoints")

- **AI Object Detection:** Detects presence/absence of expected components
- **Distance estimation:** Estimates distance between expected object location and detected object location
- Not a general-purpose defect classifier — appears to be template-matching or pre-trained object detection (likely YOLO-family or similar) for specific component types
- Integration described as "easily integrates various inspection modules"

Critical analysis: The AI module is positioned as an add-on, not the core. The vague description suggests early-stage capability — useful for binary OK/NG detection on known component types, but unlikely to match dedicated vision inspection systems (like VisionKing) for novel defect classification.

4.4 Measurement Capabilities

CAD-to-World Measurement (introduced 2025/2026):

- Distance estimation between CAD coordinates and real-world coordinates
- Supports assembly verification, positioning checks, tolerance evaluation

Critical analysis: "Distance estimation" is the operative word. Consumer-grade AR on tablets/phones typically achieves **5–15mm accuracy at best** in controlled conditions (academic literature on ARKit/ARCore). This is adequate for detecting a 20mm misalignment but wholly insufficient for precision tolerances (±0.1mm). The product carefully avoids publishing accuracy numbers, which is telling.

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5. Inspection Workflow

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┌──────────────────┐    ┌──────────────────┐    ┌──────────────────┐
│  SuPAR Composer  │    │    SuPAR App     │    │  SuPAR Web Viewer│
│  (PC - Offline)  │──►│  (Mobile - Field) │──►│  (Cloud - Review)│
│                  │    │                  │    │                  │
│ 1. Import CAD    │    │ 4. Point at part  │    │ 7. View 3D report│
│ 2. Tessellate    │    │ 5. AR snap-on     │    │ 8. Annotate      │
│    edges         │    │ 6. Inspect points │    │ 9. Share         │
│ 3. Define        │    │    - OK/NG mark   │    │                  │
│    checkpoints   │    │    - Photo+note   │    │                  │
│                  │    │    - AI check     │    │                  │
│                  │    │ 7. Generate report│    │                  │
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Guided inspection flow:

1. Inspector opens pre-configured project on tablet/phone
2. Points camera at physical part
3. AR overlay "snaps" onto real geometry via edge matching
4. Color-coded checkpoints guide systematic inspection (red/green)
5. Inspector marks each point: pass/fail + photo + comments
6. System generates report automatically (PDF/Excel/DOCX/3D)
7. Report uploaded or emailed to QA system

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6. Target Applications

Application	Suitability	Notes
Incoming goods inspection	High	Quick presence/absence check of sheet metal prototypes
Fixture/jig assembly verification	High	Verify component placement against CAD layout
Tool quality check	Medium	Visual-only; cannot replace dimensional measurement
Gripper geometry assessment	Medium	Good for gross deformation detection
Assembly line validation	High	Guided AR checklist for complex assemblies
Weld seam / sealant path inspection	Medium	Polyline markup feature; visual only, no penetration testing
Dimensional inspection (tight tolerance)	**Low**	No published accuracy; cannot replace CMM/scanner
Surface defect detection (scratches, dents)	**Low**	Not designed for surface defect classification

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7. Integration & Ecosystem

Integration	Detail
Microsoft Teams	Built-in video conferencing for remote expert guidance
Report formats	PDF, Excel, DOCX, 3D (proprietary viewer format)
Cloud storage	Projects can be opened from cloud storage
Database interface	Communication notes can be stored in database
Offline mode	Supports field work without connectivity
CAD formats	**Not disclosed**

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8. Pricing Model

Component	Price
SuPAR App	Free download (iOS/Android)
Standard tier (in-app)	$249.99
SuPAR Composer	Not publicly disclosed (likely enterprise quote)
Enterprise licensing	Contact vendor

The $249.99 app-level purchase is unusually affordable for industrial AR software. This suggests either a freemium model (basic inspection free, advanced features behind paywall) or that the real revenue comes from Composer licenses and enterprise deployments.

9. Competitive Landscape

Platform	Developer	Key Differentiator	Compared to SuPAR
Vuforia Studio	PTC	Deep PLM integration (Windchill), enterprise scale	More powerful but far more expensive and complex
Twyn	Visometry	Assisted Deviation Detection (auto-highlight discrepancies)	Most direct competitor; more quantitative deviation analysis
DELMIA Augmented Experience	Dassault Systèmes	CATIA/3DEXPERIENCE integration	Enterprise-heavy, requires Dassault ecosystem
Manifest	Taqtile	HoloLens-native, hands-free inspection	Headset-based (vs tablet-based SuPAR)
HoloCode	HoloCode	Spatial computing + digital twin	More IoT-focused
AugmentedPro	AugmentedPro	Procedure-based Creator/Player model	More workflow-oriented
FactoryOptix	AIS	AI inspection + AR overlay	More AI-heavy
Tecnomatix	Siemens	Full manufacturing simulation suite	Industrial-scale, overkill for simple visual checks

SuPAR's positioning: Lightweight, mobile-first, affordable entry point for AR visual inspection. Lower barrier to entry than enterprise solutions (PTC, Dassault, Siemens) but less powerful. Closest competitor is **Twyn by Visometry**, which offers similar CAD overlay functionality with more explicit deviation detection.

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10. Critical Assessment

Strengths

1. **Low barrier to entry** — Free app download, $249.99 upgrade, runs on existing iPads/phones
2. **No special hardware** — Consumer devices vs. expensive scanners or headsets
3. **Three-component architecture** — Clean separation of preparation, inspection, and review
4. **Fast deployment** — Minimal training needed; guided inspection workflow
5. **75% speed improvement** is credible for visual go/no-go checks vs. paper drawings
6. **Offline capability** — Critical for factory floor environments with poor connectivity
7. **Growing AI module** — Object detection adds value beyond manual visual comparison
8. **Free Web Viewer** — Lowers friction for report distribution across organization

Weaknesses

1. **No published accuracy** — The single most critical gap. No tolerance, no repeatability data, no comparison to ground truth. For a tool sold alongside ZEISS/GOM metrology equipment, this is conspicuous

2. **CAD format support undisclosed** — Industrial users need to know if their CATIA V5, NX, Creo, or STEP files are supported before evaluating
3. **Consumer hardware limitations** — iPad/phone cameras and IMUs fundamentally limit achievable alignment accuracy. Lighting, reflections, and part surface finish will affect tracking
4. **No quantitative deviation output** — Unlike Twyn (which auto-highlights deviations), SuPAR appears primarily qualitative (visual overlay + human judgment)
5. **AI module vaguely described** — "Easily integrates various inspection modules" is marketing language, not a technical specification
6. **Limited user reviews** — Apple App Store shows "not enough ratings"; suggests limited adoption or niche market
7. **Developer transparency** — CDMVision's website lacks technical documentation, white papers, or peer-reviewed validation
8. **Android parity unclear** — Most demos show iPad; Android feature parity not confirmed

Opportunities (for Strokmatic context)

- SuPAR could be evaluated as a **lightweight incoming inspection tool** for sheet metal parts (DieMaster use case)
- The AR overlay concept could complement VisionKing's automated surface inspection — SuPAR for manual assembly checks, VisionKing for defect classification
- CDMVision's **CAD2AR Framework** might be licensable for custom AR applications

Risks

- Vendor is a relatively small Turkish company — enterprise support and long-term viability should be evaluated
- No published validation against ground truth measurements makes it difficult to qualify for regulated industries (aerospace, automotive IATF 16949)
- The $249.99 price point may indicate immature monetization — risk of future pricing changes or feature gating

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11. Technical Gaps & Unanswered Questions

Question	Status
What CAD formats does Composer import?	**Undisclosed**
What is the alignment accuracy (mm) under controlled conditions?	**Undisclosed**
What is the tracking repeatability?	**Undisclosed**
Does the AI module use pre-trained or custom models?	**Undisclosed**
Is ARKit/ARCore the underlying AR framework?	**Probable but unconfirmed**
What is the Composer system requirement (OS, RAM, GPU)?	**Undisclosed**
Is there an API/SDK for custom integration?	**Undisclosed**
What is the enterprise license cost for Composer?	**Undisclosed**
How does tracking perform on reflective/dark/featureless surfaces?	**Undisclosed**
Is there multi-user concurrent inspection support?	**Undisclosed**

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12. Conclusion

SuPAR is a **well-packaged visual inspection aid** — not a metrology instrument. It solves a real problem (comparing physical parts against CAD in the field, quickly) at an accessible price point. The three-component architecture (prepare → inspect → review) is clean, and the mobile-first approach removes hardware barriers.

However, the near-total absence of quantified technical specifications (accuracy, supported formats, system requirements) makes formal qualification impossible without hands-on evaluation. For any application where tolerances matter, SuPAR should be treated as a **screening tool** (fast go/no-go) followed by proper measurement with 3D scanners or CMMs.

For Strokmatic's industrial context, SuPAR is worth monitoring as a potential complement to the existing product line, particularly for DieMaster's incoming part inspection workflows where speed matters more than micrometer-level accuracy.

Sources

- Trilion — SuPAR Product Page
- SuPAR Official Site (supar.eu)
- CDMVision Company Page
- Europac3D — SuPAR Reseller Page
- Metrology News — AR and AI Transform Visual Inspection
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- Google Play — SuPAR
- Trilion Blog — Revolutionizing Inspection
- HoloCode — Recommended Industrial AR Platforms
- FABTECH Expo — SuPAR Listing
- APM Technologies — SuPAR